

When Is Intelligence Worth Spending?

Epure Arena and the Coordination Frontier

Christophe Pennetier · 2026 · note version 0.2

Abstract. Modern systems must decide not only *what* action to take, but how much intelligence to spend producing it: a rule, random search plus verification, a learned proposer, a solver, or a human. Yet most benchmarks score only the final action; they cannot tell whether the extra intelligence was necessary or whether it paid for itself. We introduce Epure Arena, an eval-blind decision environment built on Ephemeris Kernel, with exact counterfactual pricing, deterministic replay, and small-scale oracle regret. The arena separates proposal generation from value estimation: a policy cannot see the answer key, while every candidate is scored exactly. Across four controlled regimes and a planted coordination family, we find that learned proposal generation has a narrow operating envelope. It adds value only when improvements require coordinated moves, random discovery is inefficient but feasible, and partial proposals provide informative graded feedback. When good moves are abundant, random sampling plus verification is sufficient; when they are too rare, all cheap methods starve; and on all-or-nothing constructions, adaptive refitting underperforms random exploration. We call the useful band the coordination frontier. The contribution is not a new optimizer, but a certified apparatus for measuring when additional intelligence earns its cost.

I built a test track for decision intelligence. Modern systems can spend a cheap rule, a learned model, search, simulation, a solver, a verifier, or a human on the same decision — but they rarely know whether the extra intelligence was necessary. Epure Arena makes that question measurable: it hides the answer from the policy, prices proposed decisions exactly, and measures regret against a certified optimum at small scale. Its first result is the **coordination frontier** — a narrow regime where learned proposal generation earns its cost, and outside of which cheaper methods win. The result motivates **Cenacle**: an agentic decision architecture that diagnoses the regime first, then spends the cheapest sufficient intelligence.

Reading note. A research note, not a full paper. Every number is quoted from a committed experiment ledger and carries its scope. The central positive result is an *existence* result at stated scope; the central *contribution* is the environment and the map it produces, not any single learned method. Claims carry labels — **PROVEN** | **CHARACTERIZATION** | **FRAMING** — where the distinction matters.

1. The decision above the decision

Most work on decision-making asks *what is the best action?* A growing class of systems must first answer a harder question: *which kind of intelligence should produce the action at all?* A cheap rule, a random proposal filtered by a verifier, a learned proposer, a bounded solver, a human — each has a cost, and the right choice is not fixed. This is already visible wherever compute is routed selectively: mixture-of-experts gates activate only some experts per input (Shazeer et al., 2017); LLM-cascade systems such as FrugalGPT (Chen et al., 2023) and learned routers such as RouteLLM (Ong et al., 2024) decide when a stronger, costlier model is worth invoking. In all of these the scarce resource is no longer only capacity — it is *intelligence itself*, and knowing when to spend it is the lever.

The principle we want to make measurable is simple to state and hard to verify: **use the cheapest sufficient intelligence, and prove that any escalation paid for itself.** Most benchmarks cannot check the second clause — they score the final action, not the return on the intelligence that produced it. We study the question in a setting where it *can* be checked exactly: hard constrained-decision problems where global optimization is intractable, a better value *estimate* is often irrelevant (the missing value is not a scoring error but an action the cheap policy cannot express), and the natural role for learning is not as a scorer but as a *proposer* of coordinated moves that a verifier certifies.

2. Epure Arena

Epure Arena is a decision environment built on a neutral discrete-event runtime — **Ephemeris Kernel**: an integer-time event queue with a total event order, deterministic replay, append-only event and state ledgers with byte-identical snapshots, and cryptographic run manifests. The kernel knows nothing of decisions; the arena adds four things that make the *return on intelligence* measurable rather than assumed.

- **An exact counterfactual verifier.** Given any proposed coordinated move — a bundle of admissions, drops, and re-routings — the verifier returns its exact certified value delta by separable pricing, in time linear in the size of the move, with no simulator rollout. The reward is ground truth, not an estimate. The pricing identity is *algebraically exact*; implementations agree to relative error below 10^{-13} . **| PROVEN**
- **A small-scale optimal oracle.** At $n = 12$, an ILP computes the exact optimum *over the arena's declared action space*, so any policy's regret is measured against ground truth, not a heuristic reference.
- **An eval-blind wall.** The policy structurally cannot read the answer key; the oracle is used only to compute the regret denominator, never inside candidate generation. Non-circularity is attested at the level of imports and symbols, not comments — a proposer that reached the answer through the oracle would be worthless, and the wall makes that structurally impossible.
- **A separation of proposal from valuation.** Candidate *generation* and value *estimation* are distinct layers. This is what lets us ask whether the missing value is a scoring problem (fix the estimate) or a vocabulary problem (generate a move the cheap policy cannot express).

The central metric is **coordination capture**,

$$\varphi = \frac{V_{\text{policy}} - V_{\text{greedy}}}{V_{\text{oracle}} - V_{\text{greedy}}},$$

the fraction of the greedy-to-optimal gap a policy recovers: $\varphi = 0$ is no improvement over the cheap baseline, $\varphi = 1$ is the certified optimum. Every result below is a φ measured against a certified ceiling.

The probe we use to test whether certified feedback contains *learnable* signal is deliberately minimal — a cross-entropy method (CEM) that samples coordinated bundles, prices them exactly, keeps the elites, and refits its sampling distribution toward them, round over round (Rubinstein & Kroese, 2004). It is a distribution-shifter, not a solver: it emits hints, and an exact non-learned layer owns all feasibility and pricing. Because the verifier is exact, there is no learned value critic and no bootstrapping — the only learned object is the proposal distribution. We use *learned* in the narrow operational sense that the proposal distribution is *updated from verifier feedback*; CEM is adaptive stochastic search, not a gradient-trained neural model. **CEM is a thermometer for learnable structure, not the product.** The product is the arena and the map.

3. Three terms, defined once

The map below turns on three properties of a decision problem. We define them precisely here so the rest of the note can use them without qualification.

- **Coordination-bound.** Value exists above the baseline, but no allowed local move (single-item, pairwise) or fixed-menu choice reaches it; improvement requires a *jointly constructed multi-item move*.
- **Scarce-but-reachable.** Positive proposals are *rare* under random sampling, but *discoverable* within the measured verifier budget. (Extremely scarce value — undiscoverable at any affordable budget — is a distinct, *starved* regime.)
- **Graded.** Incomplete proposals produce *ordered intermediate certified values*, so adaptive sampling can distinguish warmer candidates from colder ones. Its opposite is an *all-or-nothing* landscape, where any incomplete attempt scores like no attempt.

4. The coordination frontier: the map

The central result is a decision map. Two structural axes — coordination-boundness and informative partial credit — determine which *kind* of intelligence is the correct spend; a third, scarcity, governs whether any affordable method can find value at all.

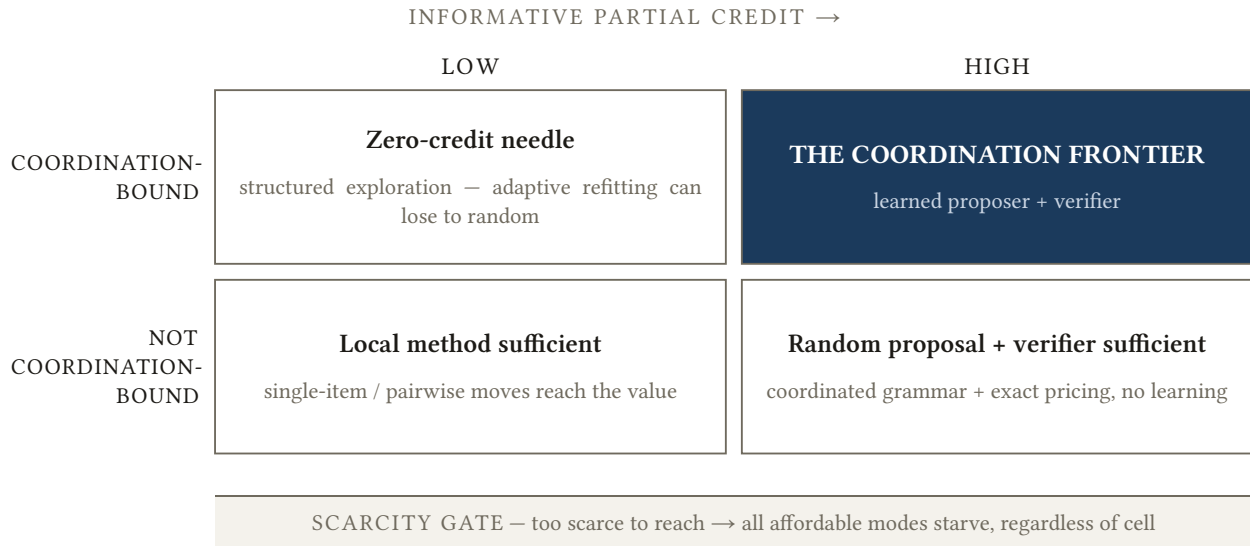


Figure 1: The decision map: two structural axes select the correct intelligence to spend; scarcity gates whether any affordable method reaches value at all. The frontier cell is the note’s result; the other three cells are its controls.

Stated as a rule: **spend a learned proposer only where the problem is coordination-bound, scarce-but-reachable, and graded — the frontier cell. Among the tested decision modes, a cheaper method is sufficient outside the frontier cell.** The verifier supplies truth; the learner supplies taste; and taste is worth paying for only where value is scarce and graded. The rest of this note is the evidence that places each regime in a cell — and one controlled construction that shows coordination-boundness alone is *not* enough to justify learning.

The bridge from this cell to the operating decisions that motivate it — which model to call, when to escalate, when to invoke a solver — is stated as a hypothesis, not a claim, in §8.

5. Evidence I — the operating-envelope map across four regimes

We tested, across four pre-registered regimes, whether a learned proposer recovers coordination value that simpler methods cannot. Three conditions must all hold for learning to be worth spending: **reachable** (some non-oracle-seeded method crosses $\varphi > 0$), **coordination-bound** (single-move, pairwise, menu-selection, and polish stay at zero), and **learning-separates** (the learned proposer beats random under the identical grammar and budget). Reading the results into the map:

regime	reachability	local	learning gain	reading
sparse enforced	rare	dead	yes, on subset	frontier candidate
grid	high	nonzero	none	random + verify
advisory	moderate	small	none material	grammar sufficient
prospective screen	unstable	nonzero	inconclusive	selector failed

Per-arm coordination capture — every arm’s mean and 95% CI — is reported in Figure 2; the ledger filenames live in the reproduce map (appendix).

The shape is legible. Where the residual is *hard to reach* (backbone-enforced), only a minority of worlds are reachable at all, but on those worlds nothing simpler than learning crosses zero. Where the residual is *easy to reach* (grid, advisory), a coordinated grammar with an exact verifier — even random sampling — already captures most of the value (random reaches 80–83% of CEM’s capture), so learning is unnecessary. **Reachability and learning-necessity are anti-correlated.** | CHARACTERIZATION

A note on the backbone-enforced verdict. Its pre-registered reliability gate — an aggregate confidence interval excluding zero — was *not met* (missed by 0.016). The same run nevertheless contains a positive per-world existence signal (§6). These are not in tension: the aggregate reliability claim is not made; the per-world existence claim is. Figure 2 reports each regime by its *decision* (rarely reachable; learning unnecessary; grammar sufficient; prospective selector failed), with the pre-registered GO/NO-GO capsules kept in the reproduce appendix.

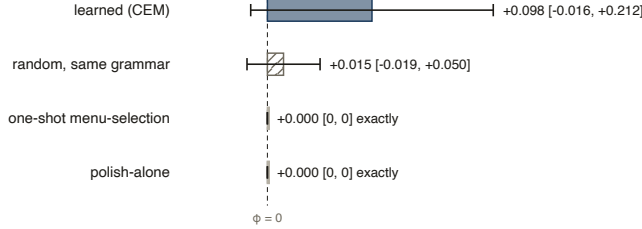
The motivating negative, in one line. An earlier proposer restricted to one-shot *menu-selection* — choose, per unit, from a fixed candidate list — recovered exactly zero ($\varphi = +0.000$, 10/10 worlds), because the useful value lives in route *combinations absent from any per-unit menu*. This isolated the bottleneck as candidate *generation*, not value *estimation*, and motivated the shift from selection to distribution-shift.

Certified capture by arm — four pre-registered regimes

coordination capture ϕ : bars = means, whiskers = 95% CI · $n=10$ worlds per run · identical grammar, verifier, and budget across arms

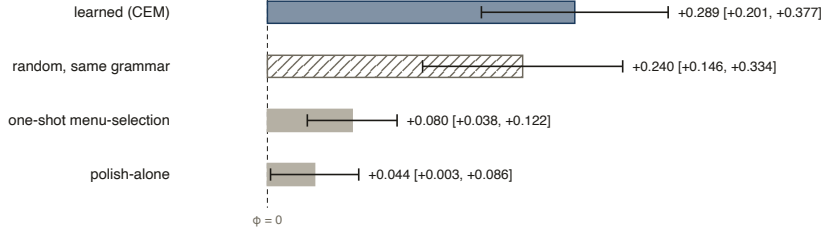
run #1 — backbone-enforced (hard, sparse)

Rarely reachable — existence signal present



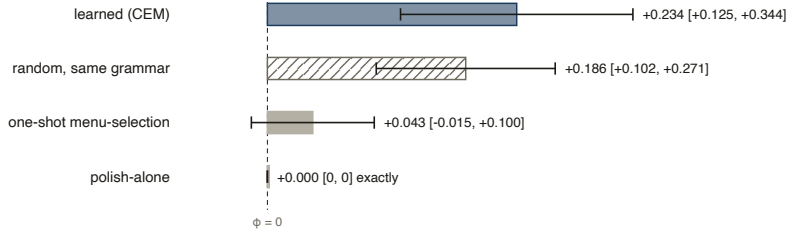
run #2 — grid (abundant)

Learning unnecessary — random already reaches it



run #3 — backbone-advisory (mid)

Grammar sufficient — no material separation



run #4 — screened, prospective

Prospective selector failed

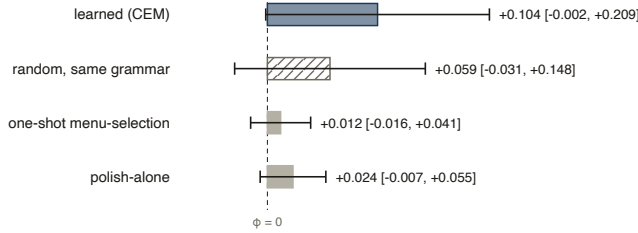


Figure 2: Certified capture by arm across the four pre-registered regimes. Bars = means, whiskers = 95% CIs; the learned arm is solid, the random arm hatched; capsules give each regime’s decision reading. · Reproduce: F2-A-F2-D (appendix).

6. Evidence II — existence, and reachability as a curve

Existence. On 3 of the 10 pre-registered worlds in the hard regime, the learned proposer is certified-positive. The three worlds show *two distinct mechanisms*, and the distinction is more informative than a single headline (Figure 3): on world 241, **amplification** — learning improves on value random already reaches (+0.284 vs +0.152); on worlds 243 and 245, **discovery** — learning finds value random does not (+0.344 and +0.355 against exact and floating-point zeros).

On two of the three worlds, random sampling with the *identical* grammar, verifier, and budget found nothing — only the arm that refits toward certified elites crossed zero. On the third, learning *amplified* value random

already reached. This is a clean mechanism isolation: same grammar, same verifier, same compute, and distribution-shift is what separates.

Scope, stated plainly. This is 3 of 10 worlds in one regime, at one budget — a *retrospective* subset, selected after the run because it was existence-positive, not a prospective population. It is evidence that the mechanism is *real*, not that it is *reliable*. | CHARACTERIZATION — *existence at stated scope*.

Three certified-positive worlds: one amplification, two learned-only discoveries

run #1, backbone-enforced · per-world certified ϕ (bars; single-run point values) · same grammar, verifier, and budget in every arm

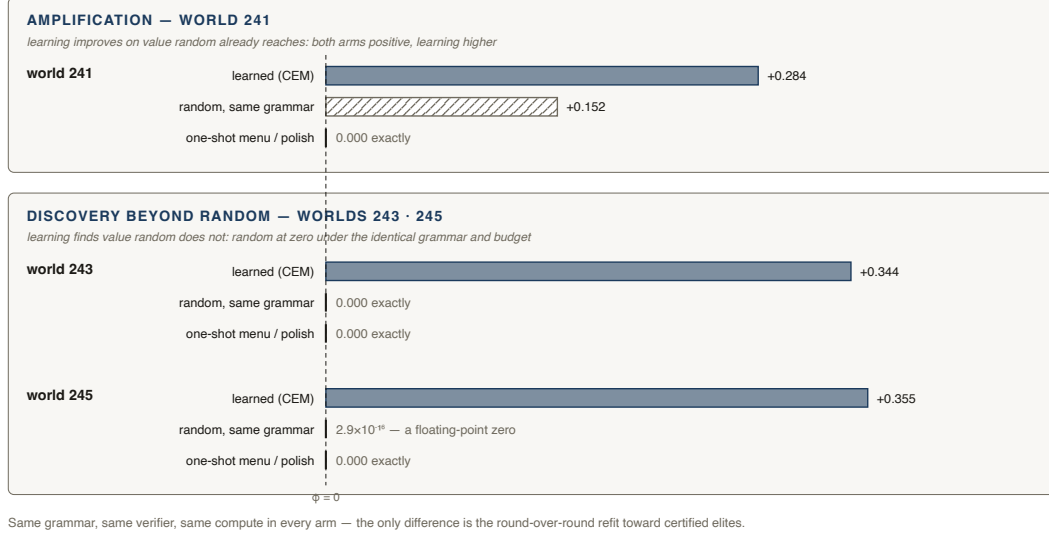


Figure 3: The three certified-positive worlds of run #1, grouped by mechanism: one amplification, two learned-only discoveries. Bars = per-world certified ϕ (single-run point values). · Reproduce: F3 (appendix).

Reachability as a curve. Fixing a single budget hides the mechanism. Sweeping the verifier budget $K \in \{25, 50, 100, 200, 400, 800\}$ (a doubling scale) and measuring $P(\phi > 0.02)$ across 93 worlds makes the envelope quantitative:

- **existence subset (retrospective, $n = 3$).** The CEM curve shifts *left* of the random curve under the same grammar and budget — CEM reaches a given probability at no more budget than random, and less at most budgets tested (at $K = 800$, 0.30 vs 0.20). The sweep *shows* a consistent directional shift within this selected subset; it does not establish a population-level effect, and “certified” is reserved for the verifier’s exact value claims, not for this descriptive curve. This panel is descriptive, not population-level.
- **grid (abundant, 30 worlds).** CEM and random curves are effectively identical at every budget — learning buys nothing where value is this reachable.
- **backbone-advisory (mid, 30).** Near-identical — the grammar plus exact verifier does the work, not the learning.
- **backbone-enforced, unscreened sparse (30).** Both curves flat-low across the whole sweep — sparsity starves every arm; the small CEM edge never clears materiality.

The left-shift on the existence subset is the entire positive learning signal, made into a curve. It appears in exactly one of four regimes.

7. Evidence III — the controlled anchor, and its refutation

To isolate the mechanism from sampling luck, we *construct* 40 worlds with a planted alternating k -cycle (per-step value Δ , coordination order $k \in \{3, 4, 5, 6\}$) so that coordination-boundness is built in. Both construction properties are checked exhaustively: 1-opt and 2-opt are exactly neutral on **40/40** worlds, and the cycle lift equals $k \cdot \Delta$ to numerical tolerance on **40/40** worlds. | PROVEN

The expected half of the result holds: reachability falls sharply as k rises in *both* arms (abundant at $k = 3$, essentially unreachable at $k = 6$) — coordination difficulty scales with order, as designed.

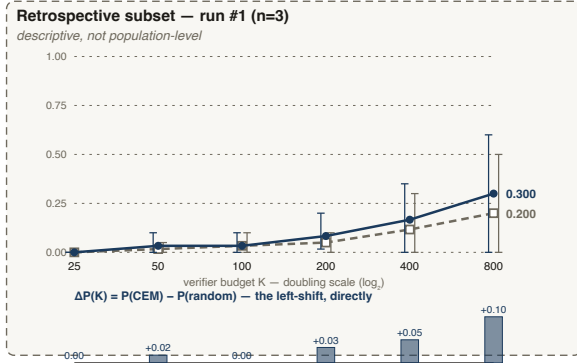
The pre-registered expectation — that learning would separate at medium k — is refuted. Random beats CEM at every k ($P(\text{find})$ at $K = 800$: $k = 3$, 1.00 vs 0.89; $k = 4$, 0.80 vs 0.47; $k = 5$, 0.28 vs 0.16; $k =$

Reachability is a curve: $P(\text{certified } \phi > 0.02)$ vs verifier budget K

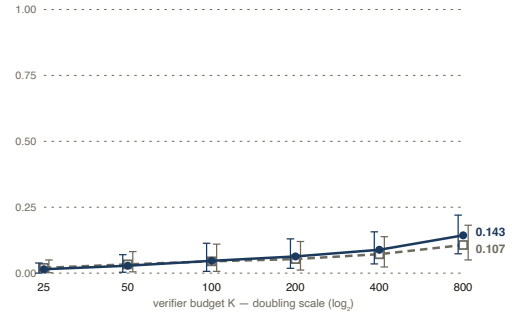
K axis is a doubling (log-2) scale · 20 proposal-RNG seeds per world per budget · whiskers = bootstrap 95% bands over worlds ($B=1000$) · $\varepsilon=0.02$ a priori

—●— learned (CEM) — solid, filled circle — □ — random — dashed, open square

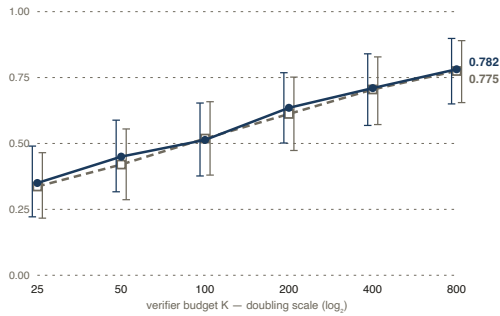
┆ bootstrap 95% band (whisker per budget)



backbone-enforced, unscreened sparse (30 worlds)



grid, abundant (30 worlds)



backbone-advisory, mid (30 worlds)

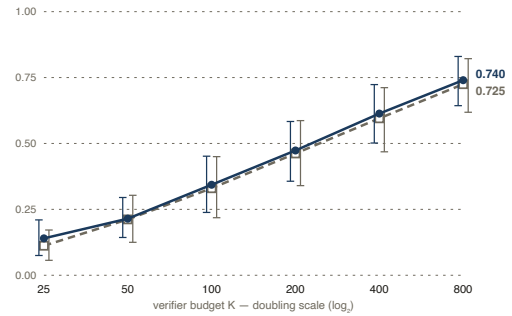


Figure 4: $P(\text{certified } \phi > 0.02)$ vs verifier budget K (doubling scale). Lines = point estimates, whiskers = bootstrap 95% bands; the retrospective run-#1 subset is segregated (tinted, dashed border) and carries a $\Delta P(K)$ inset showing the left-shift directly. · Reproduce: F4-A–F4-D (appendix).

6, 0.09 vs 0.04), with a *growing relative advantage* for random ($1.1\times$ at $k=3$ to $2.3\times$ at $k=6$; the absolute gap peaks at $k=4$). We report it as refuted.

The mechanism is clean and it *sharpens* the whole result. A pure alternating cycle is an all-or-nothing needle: any bundle completing fewer than k of the k required swaps scores at or below zero — **no partial credit**. So the elites CEM selects carry no informative ranking signal (an incomplete attempt scores like no attempt), and the refit concentrates the sampler’s mass *away* from the answer. Random, which never refits, keeps exploring and accumulates independent draws at the needle as budget grows.

CEM is a partial-credit exploiter, not a general coordination learner. Where the certified landscape between “nothing” and “the full bundle” is graded — as in the real sparse worlds of \$6, where partial drops and re-routes carry intermediate certified value — distribution-shift has an informative elite signal to climb and separates from random. Where the landscape is a gradient-free needle, *adaptive elite refitting underperforms random exploration on this construction*. Coordination-boundness alone does not determine whether adaptive refitting helps; partial credit is the discriminating axis isolated by the controlled construction.

8. What is established, and what is hypothesis

Across the tested regimes, learned proposal generation helped only where three conditions aligned — coordination-bound, scarce-but-reachable, and graded. We state this as an observed frontier, not a universal law: the three-axis alignment is demonstrated on a three-world retrospective subset — the only tested region in which the full alignment appeared. Move any axis and the recommendation flips: abundant $\rightarrow$ learning unnecessary; coordination-bound but gradient-free $\rightarrow$ refitting can lose to exploration; coordination-bound and graded but too starved $\rightarrow$ nobody finds it reliably at cheap budgets.

Transfer hypothesis. | FRAMING Epure Arena demonstrates the mechanism in a controlled capacitated decision universe. We *hypothesize* that the same operating-envelope question appears in LLM orchestration, compute scheduling, grid control, and other constrained systems — that “which intelligence should this decision

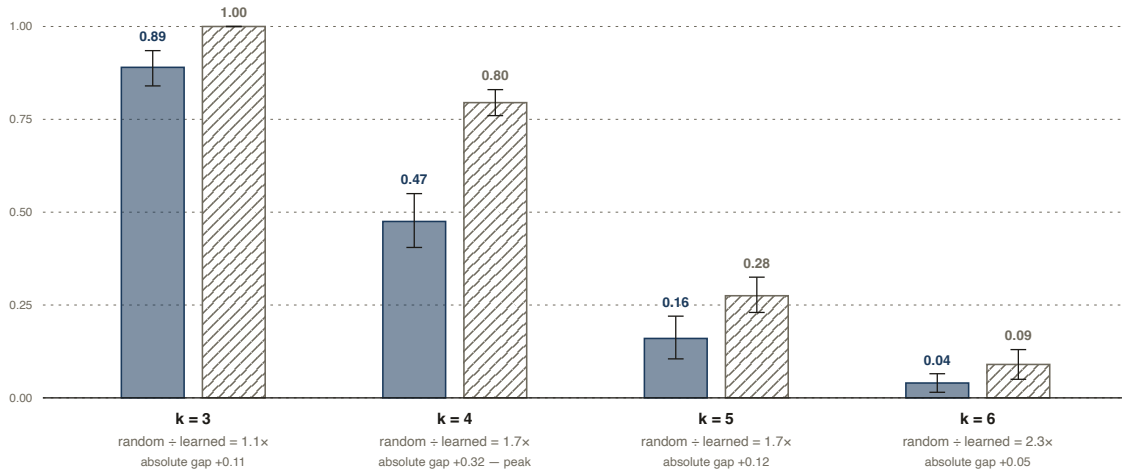
The planted k -cycle: random wins at every k — its relative advantage grows with k

$P(\text{find the planted cycle})$ at $K=800$ · 10 constructed worlds per k , 20 RNG draws per world · 1-opt/2-opt exactly neutral and cycle lift = $k \cdot \Delta$, proven exhaustively on 40/40 worlds

■ learned (CEM — refits toward elites)

▨ random (never refits)

whiskers: bootstrap 95% band



Random's relative advantage grows from 1.1x at $k=3$ to 2.3x at $k=6$; the absolute gap peaks at $k=4$. A pure alternating cycle carries no partial credit, so elites carry no informative ranking signal and the refit concentrates sampling mass away from the needle — the pre-registered separation is refuted.

Figure 5: The planted k -cycle at $K = 800$: random wins at every k with a growing relative advantage ($1.1\times \rightarrow 2.3\times$); the absolute gap peaks at $k = 4$. Bars = $P(\text{find})$, whiskers = bootstrap 95% bands. · Reproduce: F5 (appendix).

receive?” has the same four-cell structure there. These are motivations and future instantiations, **not claims established by the present experiments**. The bridge is a research direction, not a result.

9. What is open

- **A prospective selector is unsolved.** A cheap screen to select coordination-bound, graded, reachable worlds *ahead of time* failed on two axes: its hits were partly luck (4/10 did not re-fire under fresh randomness) and it selected for the wrong kind of value (admit-fill, not coordination). A refined screen is specified but not yet run. Building a *working* prospective selector — cheap observable features that predict the correct decision mode before spending — is the natural next experiment, and the first working version of a regime-diagnosing manager.
- **Small- n .** The existence result is 3/10; the prospective aggregate misses the zero-exclusion bar by 0.002 — margins small- n intervals are too wide to resolve without more worlds or a materiality-aware guard.
- **How much partial credit is enough?** We show real worlds have graded structure and pure cycles do not; we do not yet characterize the threshold of partial credit a world needs before learning separates, nor whether that threshold is measurable prospectively.

10. What comes next

The map reframes the architecture it motivates. The right system is not one optimizer nor one learned solver, but a council that *diagnoses the regime first* and then spends the cheapest sufficient intelligence: shape the situation, propose coordinated candidates, imagine their consequences, allocate a verifier budget, certify the result. This note is the foundation beneath that architecture — it establishes *why* such a system needs a regime-diagnosing manager at all, and *where* a learned proposer (or an energy-based proposal prior — a verifier-budget efficiency tool for the frontier, not a universal solver) earns its cost. The next experiment follows directly from §9: turn the retrospective map into a *prospective selector* — the first working version of that manager.

The whole evidence base was produced in roughly one workstation-hour of sub-millisecond verifier calls, every claim tied to a committed, regenerable ledger, and the one refuted prediction reported as refuted. That is the point of a certified environment: it does not only produce wins — it produces a map, and it tells you, *before you spend*, whether intelligence is worth it.

Reproduce & artifacts

Epure Arena and Ephemeris Kernel are open. The public artifact independently rederives every reported number and figure from committed ledgers; the experiment harness that *generated* those ledgers (world generation and the run driver) is presently private but named by each ledger's reproduce target. So the checkable

public chain is *ledger* $\rightarrow$ *verified numbers* $\rightarrow$ *figures*; end-to-end regeneration from raw world generation is not yet public.

- Note version 0.2 · 2026
- Epure Arena — github.com/cpennetier/epure-arena
- Ephemeris Kernel — github.com/cpennetier/ephemeris-kernel
- Reproduce — [experiments/coordination-frontier/REPRODUCE.md](https://github.com/cpennetier/experiments-coordination-frontier/blob/main/REPRODUCE.md)
- Evidence commit — e372cc3 (ledgers + verification + figure scripts)

On a fresh checkout of `experiments/coordination-frontier/`: `python3 verify_note_numbers.py` re-derives every number quoted above from the committed ledgers and exits non-zero on any mismatch; `python3 make_figures.py` regenerates all four data figures from the same ledgers. Both are standard-library-only.

References

- Rubinstein, R. Y., and Kroese, D. P. (2004). *The Cross-Entropy Method: A Unified Approach to Combinatorial Optimization, Monte-Carlo Simulation, and Machine Learning*. Springer.
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., and Dean, J. (2017). Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. *ICLR 2017*. [arXiv:1701.06538](https://arxiv.org/abs/1701.06538)
- Chen, L., Zaharia, M., and Zou, J. (2023). FrugalGPT: how to use large language models while reducing cost and improving performance. [arXiv:2305.05176](https://arxiv.org/abs/2305.05176)
- Ong, I., Almahairi, A., Wu, V., Chiang, W.-L., Wu, T., Gonzalez, J. E., Kadous, M. W., and Stoica, I. (2024). RouteLLM: learning to route LLMs with preference data. [arXiv:2406.18665](https://arxiv.org/abs/2406.18665)

Appendix — reproduce map

Every data figure and number regenerates from a committed ledger (Figure 1 is the conceptual map; its cells are instantiated by F2–F5). The pre-registered GO/NO-GO gate outcomes are recorded here (the figures carry the scientific reading; the gates are process labels).

ID	ledger (ledgers/)	regenerates via	pre-registered gate
F2-A	gonogo-cem-existence.json	make_figures.py verify_note_numbers.py	/ NO-GO (reliability CI, missed by 0.016)
F2-B	gonogo-grid-existence.json	same	REJECT-TOO-EASY
F2-C	gonogo-bbadv-existence.json	same	REJECT-TOO-EASY
F2-D	gonogo-screened-existence.json	same	REJECT
F3	gonogo-cem-existence.json (rows 241/243/245)	same	— (per-world detail of F2-A)
F4-A	curves-existence.json	same	— (descriptive subset)
F4-B	curves-bbenforced.json	same	—
F4-C	curves-grid.json	same	—
F4-D	curves-bbadvisory.json	same	—
F5	planted-kcycle.json	same	construction checks proven 40/40

All ledgers at evidence commit e372cc3; commands run from `experiments/coordination-frontier/` with stock Python 3 — no dependencies. The `reproduce_target` field inside each ledger names the private experiment harness that generated it; the public, checkable half of the chain is the ledgers themselves and the regeneration scripts beside them.

THE ARTIFACT STACK

